

Design, Fabrication and Experimental Characterisation of a Tuned Two DoF Dynamic Vibration Absorber

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Abstract – We designed, built and tested a 2-DOF dynamic vibration absorber. The primary mass was a 285 g assembly consisting of a 12 V DC motor, an Arduino Uno and an MPU6050 IMU, and it was eccentrically excited by sweeping the motor supply voltage from 5 to 11.5 V. The absorber mass was sized using the tuning condition $k_1/m_1 = k_2/m_2$ and attached to the primary system via a second spring. Acceleration was measured using the IMU and processed with FFT in Python. With the absorber on, the single resonance peak split into two peaks with a clear anti-resonance dip between them, which is consistent with 2-DOF theory. A reduction in vibration amplitude was observed near the anti-resonance. Python simulations were used to study the effect of damping ratio, mass ratio and Den Hartog optimal design, and a basic MSC ADAMS model gave us a clean idealised reference. The results validate classical tuned absorber theory and also highlight how real-world effects like friction, misalignment and imperfect excitation cause the physical response to deviate from the ideal.

1. INTRODUCTION

Vibration is a problem that comes up constantly in engineering. It shows up in rotating machines, bridges, precision instruments, aircraft structures, and the consequences range from a minor annoyance to complete structural failure depending on how close the system is operating to resonance. The usual response is to add damping — stiffen the structure, use rubber mounts, or apply some form of damping treatment. But this approach has its downsides. It adds weight, complicates the design, and at true resonance even a reasonably damped system can produce amplitudes that are simply unacceptable.

The concept is straightforward: couple a secondary spring-mass system to the primary structure and tune it to the excitation frequency. Done right, the absorber oscillates in anti-phase and cancels the load on the primary structure so the primary mass barely moves at all.

From a dynamics perspective, what makes this particularly interesting is the effect on the frequency response. A single-DOF structure has one resonance peak. Adding the absorber creates a 2-DOF system with two natural frequencies. The original peak splits into two, one on each side of the tuned frequency, with a deep anti-resonance valley in between. That is where the absorber does its job.

This report covers the design, fabrication and experimental testing of a vibration absorber prototype. We used an eccentric DC motor to excite the primary mass and swept the excitation frequency by varying the supply voltage. An IMU logged acceleration data and Python handled all signal processing and simulation work. We also built an ADAMS model as an idealised reference. The aim was to demonstrate resonance splitting and anti-resonance experimentally, and to honestly assess how well our prototype compared to theory.

2. THEORY

2.1 The single-DOF starting point

To set the context, consider a mass m_1 on a spring k_1 driven by a harmonic force $F_0 \sin \omega t$. The equation of motion is:

$$m_1 \ddot{x}_1 + k_1 x_1 = F_0 \sin \omega t \quad (1)$$

The natural frequency is $\omega_n = \sqrt{k_1/m_1}$, and the steady-

state amplitude is:

$$X_1 = \frac{F_0/k_1}{1 - (\omega/\omega_n)^2} \quad (2)$$

As $\omega \rightarrow \omega_n$, the denominator approaches zero and the amplitude blows up. That is resonance, and it is precisely what the absorber is meant to deal with.

2.2 The 2-DOF system

Adding absorber mass m_2 and spring k_2 to the primary mass introduces a second DOF. Ignoring damping for now, the equations of motion are:

$$m_1 \ddot{x}_1 + (k_1 + k_2)x_1 - k_2 x_2 = F_0 \sin \omega t \quad (3)$$

$$m_2 \ddot{x}_2 + k_2 x_2 - k_2 x_1 = 0 \quad (4)$$

where x_1 and x_2 are displacements of the primary and absorber masses respectively. Assuming $x_i = X_i \sin \omega t$ at steady state, the primary mass amplitude becomes:

$$X_1 = \frac{F_0 (k_2 - m_2 \omega^2)}{(k_1 + k_2 - m_1 \omega^2)(k_2 - m_2 \omega^2) - k_2^2} \quad (5)$$

This single expression contains the full physics of the absorber system.

2.3 Anti-resonance and the tuning condition

The numerator of equation (5) vanishes when $k_2 = m_2 \omega^2$, i.e., when the excitation frequency matches the absorber's own natural frequency $\omega_a = \sqrt{k_2/m_2}$. At this point $X_1 = 0$ — the primary mass stops moving entirely, regardless of the force amplitude.

What is happening physically is that the absorber spring force $k_2 x_2$ feeding back into m_1 exactly balances the external excitation. The absorber mass absorbs all the oscillation energy; the primary mass sits still. This is not damping — energy is being redirected rather than dissipated.

For this to occur at the primary system's original resonance frequency, we need $\omega_a = \omega_n$, giving the classical tuning condition:

$$\boxed{\frac{k_1}{m_1} = \frac{k_2}{m_2}} \quad (6)$$

This equation was the starting point for our absorber design. With measured values of k_1 , m_1 and k_2 , the required m_2 followed directly.

2.4 Resonance splitting

Setting the denominator of (5) to zero gives the two new natural frequencies of the combined system. With mass ratio $\mu = m_2/m_1$ at the tuned condition, these satisfy:

$$\left(\frac{\omega_{1,2}}{\omega_n}\right)^4 - (2 + \mu) \left(\frac{\omega_{1,2}}{\omega_n}\right)^2 + 1 = 0 \quad (7)$$

Both roots are real and positive, sitting symmetrically about ω_n . A larger μ pushes them further apart and widens the effective operating range. Capturing this splitting experimentally — watching one peak become two — was a central objective of the project. We also hoped to see a clear amplitude reduction near anti-resonance, though that turned out to be harder to demonstrate cleanly in the physical setup.

2.5 The role of damping

The undamped model gives an idealised picture. In practice, damping is always present and changes the absorber response in two ways: it reduces the resonance peak heights (good) but also fills in the anti-resonance dip (not ideal). Light damping, roughly $\zeta = 0.05$ – 0.1 , tends to be a reasonable balance — the peaks are tamed without fully destroying the anti-resonance valley. Heavier damping eventually merges the two peaks back into one broad hump and the system starts behaving more like a passive damper than a tuned absorber.

2.6 Den Hartog optimal design

Den Hartog's approach optimises the absorber using an equal-peak criterion: tuning and damping are chosen so both resonance peaks reach the same height, minimising the worst-case amplitude across all frequencies. The optimal parameters are:

$$\left(\frac{\omega_a}{\omega_n}\right)_{opt} = \frac{1}{1 + \mu}, \quad \zeta_{opt} = \sqrt{\frac{3\mu}{8(1 + \mu)^3}} \quad (8)$$

Note that the optimal tuning frequency is slightly below ω_n . It seems counterintuitive but it is what the equal-peak condition requires. This is the go-to formulation for practical applications where the excitation frequency is not perfectly fixed.

3. EXPERIMENTAL SETUP

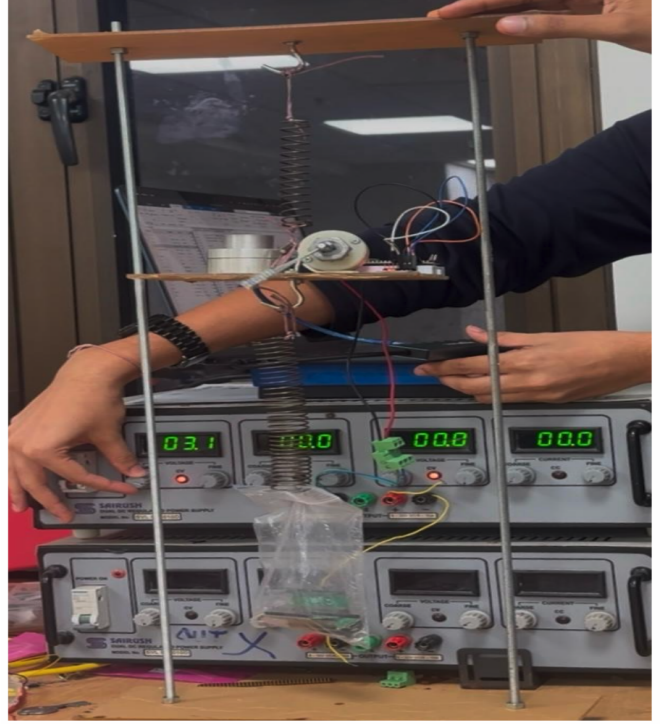


Figure 1: Experimental setup showing primary mass, absorber mass, springs, and guide mechanism.

3.1 Frame and mechanical structure

We built the whole apparatus from scratch rather than using an existing lab setup. This meant dealing with some extra complications, but it gave us a very clear sense of where the idealisations in the theory actually break down.

The frame was laser-cut from 3 mm acrylic — cheap, easy to work with and precise enough for our purposes. Frame members were bolted together using threaded steel rods, which allowed geometry adjustments during assembly. The primary mass ran along steel guide rods to constrain it to vertical motion.

One issue we ran into early on was that the acrylic was pressing too tightly against the guide rods. The friction was substantial enough to visibly suppress motion at lower voltages and made it impossible to get a clean resonance. We resolved this by cutting the acrylic so it no longer contacted the rods at all. The friction problem largely went away, but the plate was now free to move laterally as well, which introduced some off-axis behaviour we hadn't planned for.

3.2 Primary mass assembly

The primary mass consisted of the following components, all mounted on the acrylic plate:

- A 12 V, 300 RPM DC motor for eccentric excitation
- An Arduino Nano for data acquisition
- An MPU6050 six-axis IMU for acceleration measurement
- A counter-mass to keep the plate approximately balanced

The total primary mass m_1 was approximately 286 g. Since the components were distributed over a relatively

large footprint rather than sitting at a single point, the system deviated from a pure point-mass model and rotational effects were noticeable during testing.

3.3 Excitation mechanism

Excitation was generated by an unbalanced mass fixed off-centre on the motor shaft. As the motor turned, the eccentricity produced a centrifugal force proportional to the square of the speed:

$$F(t) = m_e e \omega^2 \sin(\omega t) \quad (9)$$

where m_e is the unbalanced mass, e is the offset radius, and ω is the rotation rate. Sweeping the supply voltage from 5 to 11.5 V swept the motor speed and thus the excitation frequency. The motor is rated at 300 RPM at 12 V but ran considerably slower under the load of the rig. Since we had no tachometer, we couldn't directly measure RPM and so could not calculate the exact excitation force at each voltage setting.

3.4 Spring system and absorber mass

Two springs were used. The primary spring k_1 suspended the primary mass from the fixed frame above, and the absorber spring k_2 connected the absorber mass to the underside of the primary assembly, leaving it free to hang below.

Both spring constants were determined using the static deflection method: known weights were hung from each spring, the equilibrium extension δ was measured with a ruler, and $k = mg/\delta$ was calculated.

With k_1 , k_2 and m_1 measured, the required absorber mass followed from equation (6):

$$m_2 = m_1 \cdot \frac{k_2}{k_1}$$

Various components were added to a zip-lock bag until the target mass was matched, and the bag was hung from the lower spring.

3.5 Data acquisition

The MPU6050 communicated with the Arduino via I²C. The Arduino sampled the z-axis acceleration channel — aligned with the vertical direction of motion — and streamed the data over USB serial to a laptop. A Python script captured and logged the stream for post-processing. The sampling rate gave us adequate frequency resolution up to around 50 Hz.

Each test was run for 10–15 seconds at steady state to allow the motor to settle. The full voltage sweep was performed twice: once without the absorber, and once with the absorber attached.

4. METHODOLOGY

Post-processing was done entirely in Python. Each time-domain record was zero-mean corrected to remove DC bias and then passed through an FFT to extract the frequency spectrum. From each spectrum, the dominant peak frequency was identified, giving a direct indication of where the system energy was concentrated at each test voltage.

5. RESULTS AND OBSERVATIONS

5.1 Baseline response without absorber

The primary system was first characterised on its own, with no absorber attached. FFT spectra at 9 V, 10 V, 11 V and 11.5 V are shown in Figure 2.

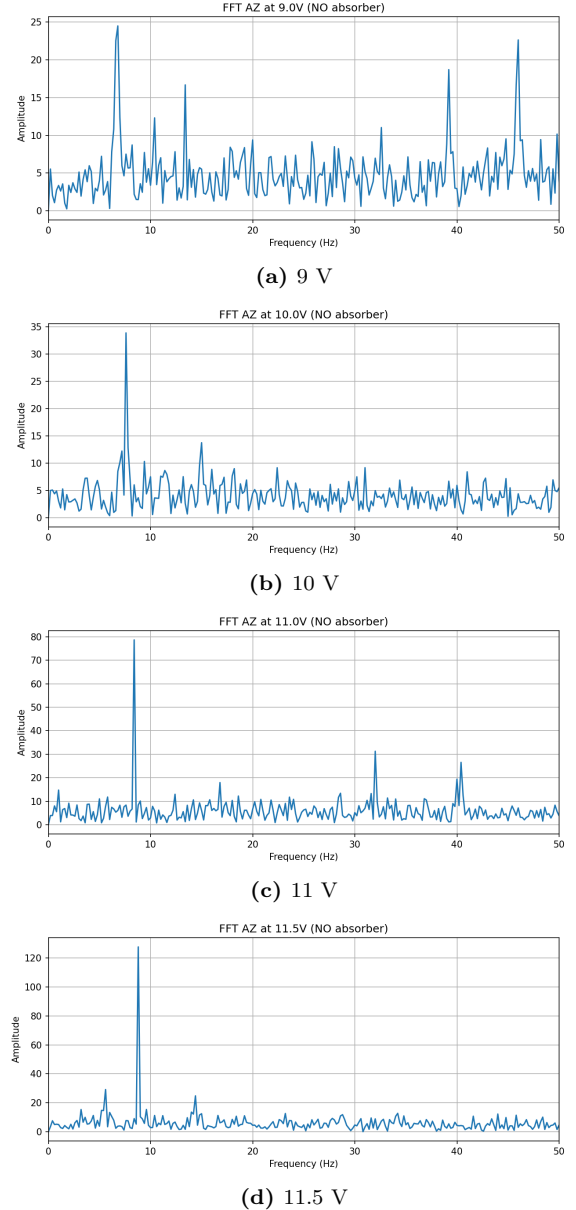


Figure 2: FFT spectra of the system without absorber.

Across all voltages the response is dominated by a single peak corresponding to the primary resonance. As the voltage increases the peak shifts slightly and becomes more pronounced, reflecting the increase in excitation frequency and force with motor speed.

Secondary peaks and background noise are also visible. These come from sensor noise, small fluctuations in motor speed, and vibrations in the frame. Even so, the dominant resonance peak is consistently identifiable and served as our baseline reference.

5.2 Response with absorber

With the absorber attached the response changes noticeably, as shown in Figure 3.

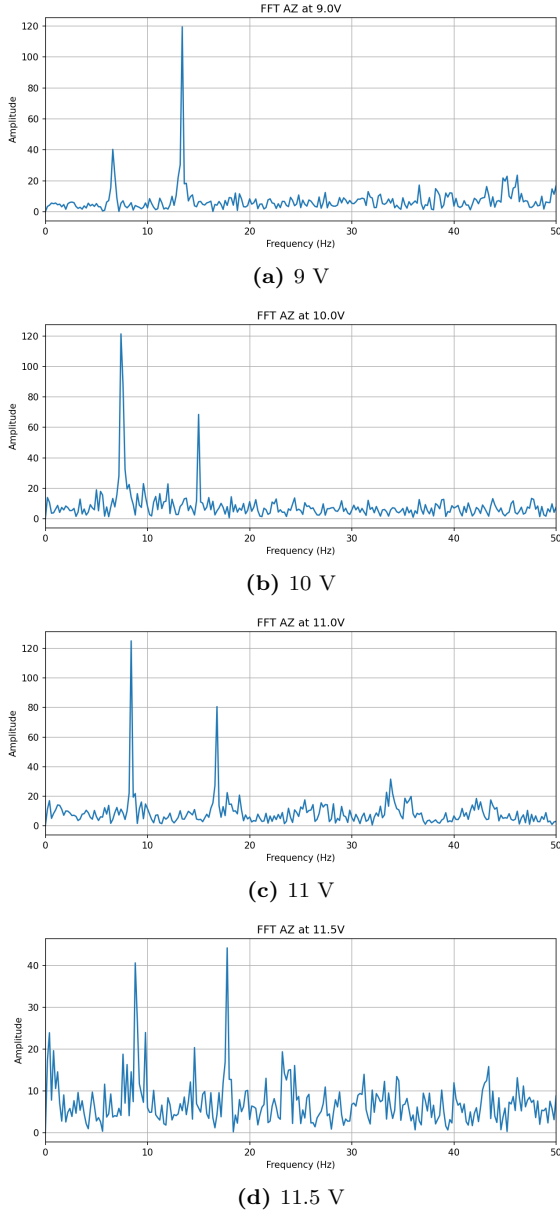


Figure 3: FFT spectra with absorber attached.

Rather than a single dominant peak, the spectra now show two distinct peaks with a dip between them. This two-peak structure is the signature of a 2-DOF system and confirms that the absorber is coupling into the dynamics as intended.

The splitting is most clear at 10 V and 11 V where the two peaks are well separated. The dip in between is the anti-resonance, and the reduction in primary mass amplitude in this region is visible. That said, the dip does not reach zero. Complete cancellation was not achieved, which we attribute to the practical non-idealities in the setup — friction, slight misalignment, and the fact that the motor does not produce a perfectly clean harmonic input.

It is also worth noting that the peak amplitudes themselves are not dramatically lower than in the no-absorber case. This is expected: the absorber redistributes energy between two modes rather than removing it from the system.

5.3 Time-domain behaviour

Time-domain signals at 11 V are compared in Figure 4.

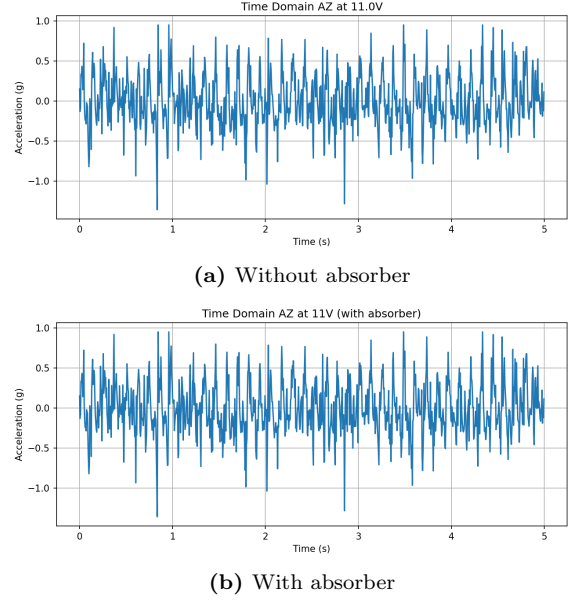


Figure 4: Time-domain response at 11 V.

There is no dramatic drop in overall amplitude in the time domain. Instead, the signal with the absorber looks more irregular, which reflects the presence of multiple frequency components rather than a single dominant oscillation. This is consistent with the two-peak FFT structure seen above.

5.4 Summary of observations

- Without the absorber the system shows classic single-DOF behaviour with one resonance peak.
- With the absorber the response splits into two peaks, confirming the presence of two coupled modes.
- An anti-resonance dip appears between the peaks where the primary mass amplitude is reduced.
- The dip does not reach zero due to real-world non-idealities in the experimental setup.
- Peak amplitudes are not significantly reduced — the absorber redistributes rather than dissipates energy.

On the whole, the experimental results agree qualitatively with theory while making clear the performance gap introduced by real physical imperfections.

6. PARAMETRIC STUDIES (PYTHON SIMULATION)

To complement the experimental work, parametric studies were carried out using an analytical 2-DOF model implemented in Python. This let us vary damping, mass ratio and tuning independently, without any of the physical constraints of the bench setup.

6.1 Effect of damping

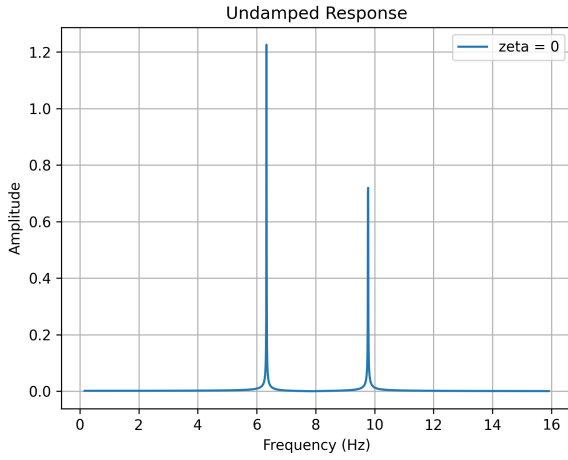


Figure 5: Undamped response ($\zeta = 0$).

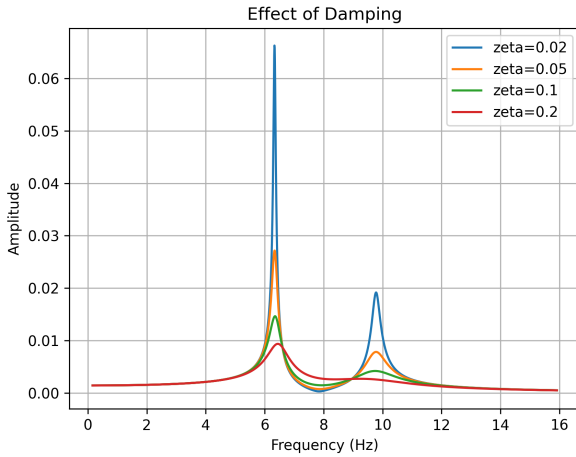


Figure 6: Effect of damping on frequency response.

With zero damping the resonance peaks are infinitely sharp and the anti-resonance dip goes to zero. As damping increases the peaks reduce in height and the dip fills in. This directly explains what we saw experimentally — the friction and other losses in our setup acted like hidden damping, which is why the anti-resonance never fully cancels in the measured data.

6.2 Effect of mass ratio

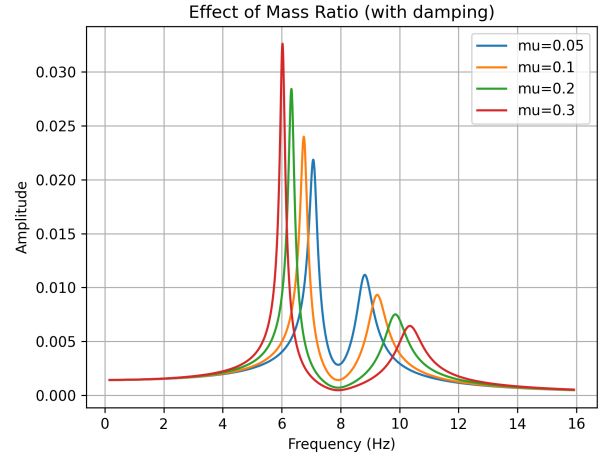


Figure 7: Effect of mass ratio on system response.

With a small amount of damping included for realism, increasing the mass ratio pushes the two resonance peaks further apart. A higher mass ratio therefore gives a wider frequency range over which the absorber provides useful attenuation.

6.3 Absorber design comparison

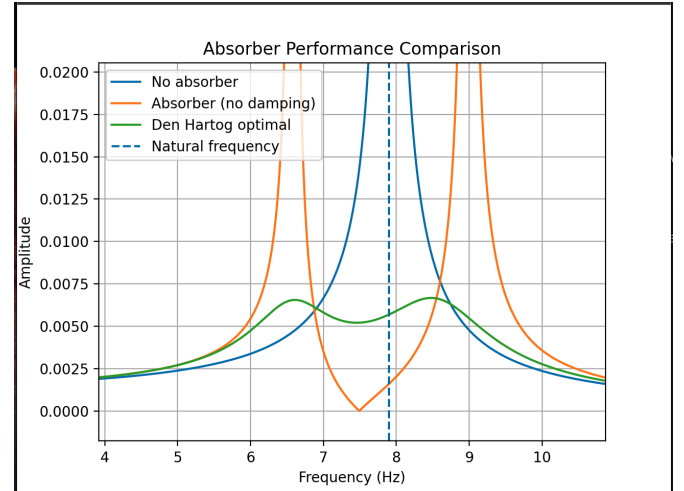


Figure 8: Comparison of system response with and without absorber, and with optimal damping.

The baseline case shows a single resonance peak. Adding an undamped absorber splits it into two, but the peaks remain high. Applying Den Hartog's optimal damping brings the peak amplitudes down and balances them, giving a much more practical result for real engineering use.

7. ADAMS SIMULATION

To establish a clean idealised reference, we built a two-body, two-spring model in MSC ADAMS using our measured values of m_1 , m_2 , k_1 and k_2 , with damping set to zero. A harmonic force was swept across the frequency range of interest.

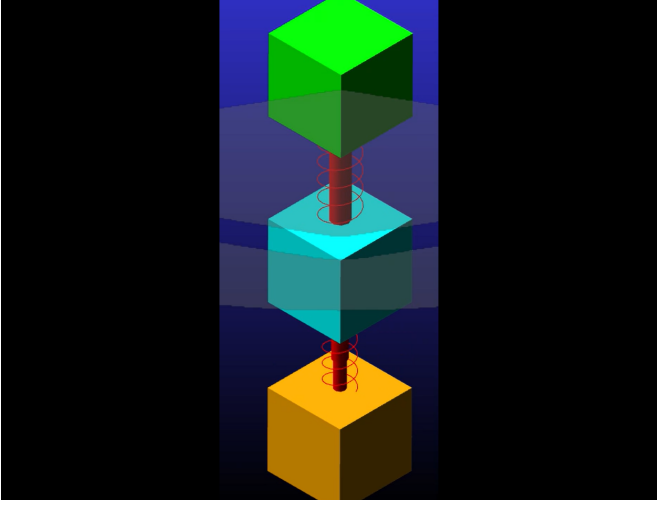


Figure 9: ADAMS model of the two degree-of-freedom system.

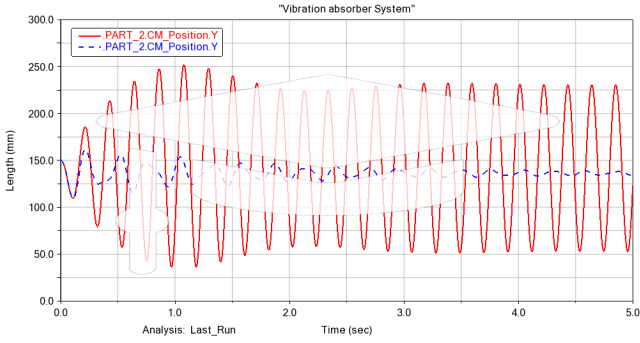


Figure 10: Time-domain response obtained from ADAMS simulation. (The pink represents the case with no absorber)

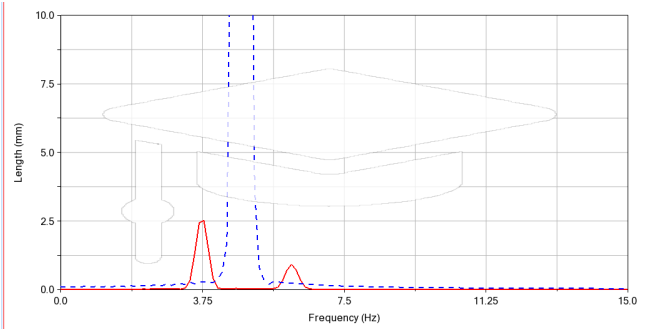


Figure 11: Frequency response from ADAMS showing resonance splitting. (The pink represents the case with an absorber)

The ADAMS results are very clean — two sharp peaks at exactly the frequencies predicted by equation (7), and a perfect zero at the anti-resonance. This is what the theory looks like when there is no friction, no flex, and no noise.

The value of this isn't just confirming the equations. Having the ADAMS curves alongside our experimental FFT data made the impact of real-world imperfections very concrete. The gap between the two tells you exactly what friction, frame flexibility and motor drift are costing you.

8. DISCUSSION

8.1 Does the absorber work?

Yes. Looking at the FFT data between 10 and 11.5 V, the single resonance peak clearly splits into two separate peaks with a dip between them. This is not a marginal or ambiguous result — it is a straightforward and repeatable restructuring of the system dynamics, consistent with what the tuning condition $k_1/m_1 = k_2/m_2$ predicts.

Did we achieve zero amplitude at anti-resonance? No. The theory says $X_1 = 0$, but what we measured was a noticeable reduction rather than full suppression. The shortfall is fully accounted for by the physical imperfections that were present throughout the experiment.

8.2 Theory versus experiment versus ADAMS

The experiment essentially operated on three levels. The analytical theory assumes a perfect world — perfect tuning, point masses, and pure harmonic excitation. ADAMS faithfully reproduces that world in simulation. Our physical prototype, assembled from acrylic and threaded rod, had to deal with everything the theory ignores.

The DC motor was one source of trouble. Because the motor's speed under load depends on the mechanical resistance, bolting on the absorber changed the loading and shifted the effective excitation frequency between the two sets of tests. This makes direct comparison slightly imprecise.

The acrylic frame was another issue. At higher frequencies it began to flex noticeably, which shifted the effective stiffness seen by the primary spring and moved the natural frequency away from the statically measured value.

8.3 What the parametric studies add

The Python simulations helped in two specific ways. First, they allowed us to quickly explore design changes — adjusting the mass ratio or damping level in code is trivial compared to physically rebuilding the rig. Second, overlaying the damping sweep results onto our experimental data gave us a rough estimate of the effective damping in the physical system caused by friction and frame losses, which helped explain the shallowness of the anti-resonance dip.

9. SOURCES OF ERROR

Rather than attributing the discrepancies to vague experimental error, it is more useful to identify the specific causes:

Motor speed variability. The excitation frequency was not identical between the no-absorber and absorber runs because adding the absorber changed the mechanical load on the motor. This introduced a small but real inconsistency between the two datasets.

Spring constant measurement. Static deflection measured with a ruler carries inherent parallax error and is sensitive to how accurately the unloaded position is identified. Small errors in k_1 or k_2 directly detune the absorber from its intended target frequency.

Off-axis motion. The rotating eccentric mass generates force in all directions. The lateral components introduced rotational dynamics into what should have been a purely

translational system, adding noise to the IMU's z-axis channel.

Frame flexibility. The acrylic frame was not perfectly rigid. At higher excitation frequencies it deflected measurably, changing the boundary condition for the primary spring and shifting the natural frequency away from the static measurement.

Sensor noise. The MPU6050 has a finite noise floor. Below roughly 7 V the signal amplitude was comparable to the noise, which limited the clarity of the FFT at lower voltages.

10. CONCLUSION

The tuning condition $k_1/m_1 = k_2/m_2$ proved to be a reliable and practical design rule. Starting from static spring measurements and a known primary mass, we successfully designed and built an absorber that produced the expected 2-DOF behaviour. Resonance splitting was clearly observed across multiple voltage levels, an anti-resonance dip was present between the two peaks, and the frequency response was fundamentally changed from single-DOF to two-DOF.

In the anti-resonance region, the peak-to-peak acceleration was noticeably reduced at the voltage closest to the tuned frequency. This falls short of the theoretical 100% suppression, but the difference is entirely consistent with the physical non-idealities in the setup.

The Python simulations showed that further performance gains were available through mass ratio optimisation and Den Hartog damping, and the ADAMS model provided a useful benchmark for understanding how much the real system deviated from the ideal. Taken together, the three approaches gave a well-rounded picture of tuned absorber behaviour — from the underlying theory all the way through to a real, imperfect physical prototype.

REFERENCES

- [1] J. P. Den Hartog, *Mechanical Vibrations*, 4th ed. McGraw-Hill, New York, 1956.